

RAJESH RAMANI

Agentic AI Engineer Fremont, CA, USA rajesh_ramani@icloud.com · (408) 806-0359 [linkedin.com/in/rvrajesh123](https://www.linkedin.com/in/rvrajesh123) · github.com/rvvenkatrama

PROFILE SUMMARY

Agentic AI Engineer building production agent systems across all three major LLM providers (Anthropic Claude, OpenAI, Google Gemini). Author of the open-source **HPRC Framework** — an AI-native templating engine with dependency-graph scheduling, asyncio concurrency, and a 6-adapter provider abstraction. Designer and builder of **MatchScout**, a gig-driven recommender with rigorous outcome-based A/B evaluation now in PoC at a creator-business marketplace startup. Track record: shipping working code with eval gates, observability, and cost tracking — not demos.

ENGINEERING CENTERPIECES

HPRC Framework — Open-Source AI-Native Templating Engine

Python · Apache-2.0 · asyncio · Pydantic v2 · FastAPI · pytest GitHub: github.com/HPRCFramework/hprc-framework · Web: hprcframework.dev

- Designed and built **HPRC** end-to-end: an open-source server-side templating engine where developers embed LLM prompts directly inside HTML (SPREP templates) and the framework executes them during page rendering.
- Declarative render pipeline:** parses templates, auto-builds a prompt dependency graph from `<include>` references, runs independent prompts **concurrently via asyncio topological scheduling**, and TTL-caches results keyed on the fully-resolved prompt. Prompt blocks never appear in output — only their responses do.
- Provider-agnostic LLM abstraction** with **six interchangeable adapters** (OpenAI · Anthropic · Gemini · Ollama/LM-Studio · Mock · MultiProvider router) behind a single coroutine interface. SDKs lazily/optionally imported so installing the framework doesn't drag every provider's SDK with it.
- External rules + tools seams** (no expression language in templates): per-prompt conditional execution and allowlisted tool invocation — sandbox-safe.
- Layered, acyclic architecture:** `models ← leaf services ← config ← renderer ← __init__`. Tolerant HTML parser, Pydantic v2 throughout.
- Authored the SPREP language specification** (mini-RFC). Full architecture docs, user guide, README, **66 passing pytest tests**, pyflakes-clean.

MatchScout — Onemyle Marketplace AI Recommender

LangGraph · Claude Sonnet 4.6 · sentence-transformers · SQLite · Streamlit · LangSmith · pytest · GitHub Actions GitHub: [Onemyle-matchscout](https://github.com/Onemyle-matchscout)

- Whole pipeline as one LangGraph state machine** (not just Stage 3): deterministic hard filter → in-memory sentence-transformer cosine ranker → arm-routed Stage 3 (Arm 1 control = top 3 by cosine, no LLM; Arm 2 = Claude re-ranks top 3 from top 10 using past-gig metrics + a versioned prompt template). Bounded retry loop on LLM validation failure (malformed JSON, wrong cardinality, hallucinated creator IDs).
- Outcome-based two-arm A/B eval:** 50% no-LLM control vs. 50% LLM treatment. Per-arm metrics with **p-value significance testing**. No synthetic golden set, no CI quality gate — quality is measured on real downstream outcomes.
- Prompt-as-tuning-surface:** versioned templates in `prompts/arm2_v1.txt`; `prompt_version` stamped on every recommendation; outcome dashboard segments metrics by version → empirical prompt iteration loop.
- Observability:** `@traceable` instrumentation on every Stage-3 LLM call ships spans to LangSmith; trace URL persisted on the recommendation row for one-click drill-down from the dashboard.
- Per-call USD cost** computed from Anthropic token usage and persisted per recommendation → cost-per-arm + cost-per-prompt-version comparison alongside quality.
- CI:** GitHub Actions pytest schema smoke test on every push (no API key — exercises the no-LLM path).

AI / ML SKILLS

- LLM APIs:** Anthropic Claude (sonnet-4.6, opus-4.7) · OpenAI GPT-4o · Google Gemini · function calling · tool_use · streaming · prompt caching · extended thinking
- Agent Frameworks:** LangGraph (StateGraph, Send API, HITL, MemorySaver, SqliteSaver) · LangChain (LCEL Runnables, ChatPromptTemplate, RunnableParallel/Branch/Passthrough/Lambda) · CrewAI (parallel + hierarchical) · Claude Agent SDK · Memo
- Protocols: Model Context Protocol (MCP)** — tools, resources, prompt templates, streamable-http; FastMCP server + agentic clients
- Agentic Patterns:** Tool-use loops · ReAct · reflection / self-critique (Reflexion) · orchestrator-worker · hub-and-spoke · multi-agent crews · subagent decomposition · arm-routed treatment design

- **RAG:** Embeddings (sentence-transformers, BERT, OpenAI text-embedding-3) · RecursiveCharacterTextSplitter separator hierarchy · token-aware chunking via tiktoken · cosine similarity · L2-normalization → dot-product optimization · citation grounding · metadata propagation through chunkers
- **Vector Stores:** ChromaDB (Ephemeral + Persistent) · Qdrant · Pinecone (familiar)
- **Memory:** Mem0 semantic memory · Zep (Graphiti temporal graphs — in progress) · checkpointing (MemorySaver, SqliteSaver) · Anthropic Memory tool · Letta (planned, P9)
- **Validation:** Pydantic v2 (BaseModel, field_validator before/after modes, model_dump_json, model_validate_json) · TypedDict for LangGraph state · structured output · fence-tolerant JSON-mode parsing
- **Async / Concurrency:** asyncio · asyncio.gather · async context managers · async generators · **dependency-graph (Kahn topo) scheduling** (HPRC) · provider-level fanout
- **Observability & Eval:** LangSmith (@traceable decorator, get_current_run_tree, span hierarchy, trace-URL drill-down from dashboards) · outcome-based A/B with p-value significance testing · per-call USD cost tracking from token usage · LangFuse (in progress)
- **Testing & CI:** pytest (fixtures, scopes, assertion rewriting, mocked provider-conformance harness) · GitHub Actions schema smoke gating · pyflakes
- **Adjacent:** FastAPI · Streamlit · SQLite · Node.js · Docker / k3s (planned, P10) · AWS Bedrock (planned, P10) · Terraform (familiar)

PROFESSIONAL EXPERIENCE

Investor / Agentic AI Product Consultant · Onemyle (Hyperlocal Creator/Business Marketplace Startup)

Mar 2026 – Present · Remote

- Lead AI strategy, technical architecture, and product direction for the startup. Currently designing and shipping **MatchScout** (see Engineering Centerpieces above).
- **Onemyle User Chatbot** — AI-powered local area intelligence assistant for Indian cities. Designed a 3-node deployment with local Gemma orchestrator, ChromaDB semantic cache, multi-LLM parallel calls (Claude · GPT-4o · Gemini), intent-based routing, and a FastAPI + React stack.
- Strategic input on roadmap, ICP refinement, AI product positioning, category taxonomy design.

Principal Program Manager · Oracle Cloud Infrastructure (OCI)

Fremont, CA · Jan 2026 – Mar 2026

- Led Fiber Supply Remediation Program for hyperscale DC builds; expanded supplier base 3 → 7; built bulk-fiber estimation tool.

Manager, Technical Program Management · Apple Inc.

Infrastructure Programs · Software Tools for PMO and DC Operations Sunnyvale, CA · Apr 2011 – Jan 2024

- Led multi-year infrastructure programs across compute, storage, network, on-prem cloud, big-data, and database environments at hyperscale.
- **Built and shipped internal tooling:** custom PMO portal and DC operations prioritization tool (DCCP) — drove **50% efficiency gain** in cross-org program tracking.
- Managed capital budgets exceeding \$20M; consolidated 400+ applications across data centers; established PCI-compliant High-Secure Environments.
- Cross-functional partners: Platform/Cloud SRE · Database Engineering · Storage · DC Ops · Apple Online Store · Apple Pay · iCloud · Applied Machine Learning.

Prior Experience (Summary)

PayPal (Senior PM, InfoSec, 2010–11) · EDS/HP (Delivery Manager, 125+ team, 2007–10) · BoA (Release Manager, 2006–07) · **iDrive Inc. (CTO, 1998–2006)** — designed and led development of a commercially successful data-backup product.

CAPSTONE PROJECTS

Built across the curriculum to demonstrate framework breadth and pattern coverage.

- **StockSage MCP Server** (FastMCP · Claude · ChromaDB · sentence-transformers) — Production MCP server with 4 stock-analysis tools over streamable-http. Generic agentic loop dispatches tools dynamically via session.call_tool(name, input) with zero hardcoding. Persistent ChromaDB vector store for earnings-transcript embeddings; MCP prompt templates decoupled from client.
- **ReviewCrew** (CrewAI · GitHub REST API · Claude) — 2-agent crew (code reviewer + security reviewer) analyzing GitHub PRs in parallel via Process.parallel. Custom BaseTool subclass for GitHub API; HITL before comment posting.
- **ResearchBot** (LangGraph · Kafka · DuckDuckGo) — Multi-step web-research agent on StateGraph with typed state, conditional edges, and MemorySaver checkpointing. HITL interrupt() / update_state() for mid-graph approval. Annotated[list, operator.add] reducer for append-only state.

- **DocTalk** (*LangChain · Qdrant · sentence-transformers · Claude*) — End-to-end RAG pipeline over PDFs: ingestion → chunking → embedding → cosine retrieval → cited answer generation. Per-page metadata propagation through `split_documents()` enables citation grounding.
 - **ToolBot** (*asyncio · Pydantic v2 · Anthropic / OpenAI / Gemini*) — Unified multi-LLM CLI agent behind a single tool-use loop. Streaming, multi-turn memory with sliding-window truncation, provider-specific dialect handling, Pydantic v2 schema validation.
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EDUCATION & CERTIFICATIONS

- M.S., Information Systems — California State University, Los Angeles
 - B.S., Computer Science — National Institute of Technology, India
 - **AWS Solutions Architect — Associate · KCNA · PMI-PMP · Certified ScrumMaster · GCP Cloud Digital Leader**
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LINKS

- LinkedIn: [linkedin.com/in/rvrajesh123](https://www.linkedin.com/in/rvrajesh123)
- GitHub org (AI projects): github.com/Agentic-and-AI-Engineering-Projects
- AI Engineering Curriculum: [ai_engineer_program](#)
- MatchScout: [Onemyle-matchscout](#)
- HPRC Framework: github.com/HPRCFramework/hprc-framework · hprcframework.dev